

# CSE 4125: Distributed Database Systems

## Chapter – 6

### (Part – C)

Optimization of Access Strategies.

# Topics to be discussed -

- Query Optimization
- Data Transmission
- Comparison between different Query Optimization Strategies

# Query Optimization

- Permutation of the ordering of operations within a query can provide many equivalent **strategies** to execute it.
- Finding an “**optimal**” ordering of operations for a given query is important.
  - Done by query optimization layer( or optimizer for short).

# Data Transmission

- Data transmission requirement can be evaluated by –
  - **Transmission cost**
    - i.e. cost to initiate a transmission, routing cost etc.
  - **Transmission delay**
    - i.e. elapse time between activation and completion of an app.

# Data Transmission (cont.)

- Data transmission requirement can be evaluated by –

- **Transmission cost**

$$TC(x) = C_0 + x * C_1$$

- **Transmission delay**

$$TD(x) = D_0 + x * D_1$$

$x$  = Transmitted data

$C$ 's and  $D$ 's are system dependent constants.

$C_0$  = initialization fixed cost

$C_1$  = network wide unit cost

$D_0$  = connection initialization fixed time

$D_1$  = network wide unit transfer rate

# Data Transmission (cont.)

- Data transmission requirement can be evaluated by *(more detailed characterization)*—

## ➤ Transmission cost

$$TC(x) = C_0^{ij} + x * C_1^{ij}$$

 Total amount of data

## ➤ Transmission delay

$$TD(x) = D_0^{ij} + x * D_1^{ij}$$

 Total / Longest amount of data (Total = Not parallel, Longest = Parallel execution)

*i and j denote source and destination respectively.*

# Comparison between different strategies

- ✓ We will see different versions of a query.
- ✓ Measure their cost and delay to see which one is better.

# Example

- You will be given a query with fragmentation schema and their database profile.
- Draw operator tree for the query.
- Apply strategy 1 as per the question
- Apply strategy 2 as per the question
- Compare between strategies



# Scenario

For the following query compare strategy 1 and strategy 2 based on the TC and TD at site 2.

**PJ**<sub>SNUM</sub> ((**PJ**<sub>SNUM, DEPTNUM</sub> **SUPPLY**) **JN**<sub>DEPTNUM=DEPTNUM</sub> (**PJ**<sub>DEPTNUM</sub>  
**SL**<sub>AREA="North"</sub> **DEPT**))

Assume that **SUPPLY** has two horizontal fragments and **DEPT** has three horizontal fragments. The database profiles of **SUPPLY**<sub>1</sub> and **SUPPLY**<sub>2</sub> are similar. Also, profiles of **DEPT**<sub>1</sub>, **DEPT**<sub>2</sub> and **DEPT**<sub>3</sub> are identical. The database profiles are shown on the next slide.

Determine the transmission delay for the above strategy when **network-wide transfer rate is 10000 bits/second**. Note that, the system **provides the benefit of parallel processing and the initial delay is 0**.

# Scenario

- Profiles of  $SUPPLY_1$  and  $SUPPLY_2$ :

$card(SUPPLY_1) = 30000$

$card(SUPPLY_2) = 20000$

$site(SUPPLY_1) = 1$

$site(SUPPLY_2) = 4$

|      | snum | pnum | deptnum | quan |
|------|------|------|---------|------|
| size | 6    | 7    | 2       | 10   |
| val  | 1800 | 1000 | 20      | 500  |

- Profiles of  $DEPT_1$ ,  $DEPT_2$  and  $DEPT_3$ :

$card(DEPT_1) = 10$

$card(DEPT_2) = card(DEPT_3) = 20$

$site(DEPT_1) = 2$

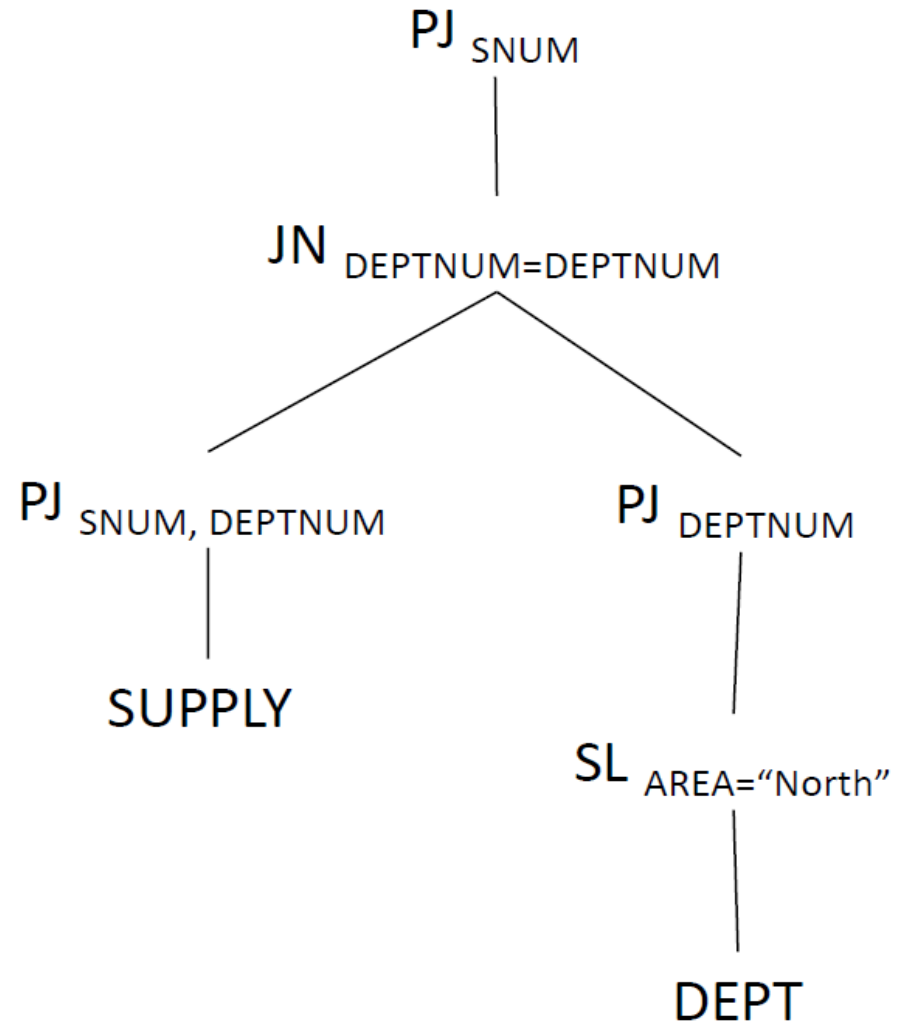
$site(DEPT_2) = 3$

$site(DEPT_3) = 5$

|      | deptnum | name | area | mgrnum |
|------|---------|------|------|--------|
| size | 2       | 15   | 1    | 7      |
| val  | 10      | 10   | 2    | 10     |

# Given Scenario

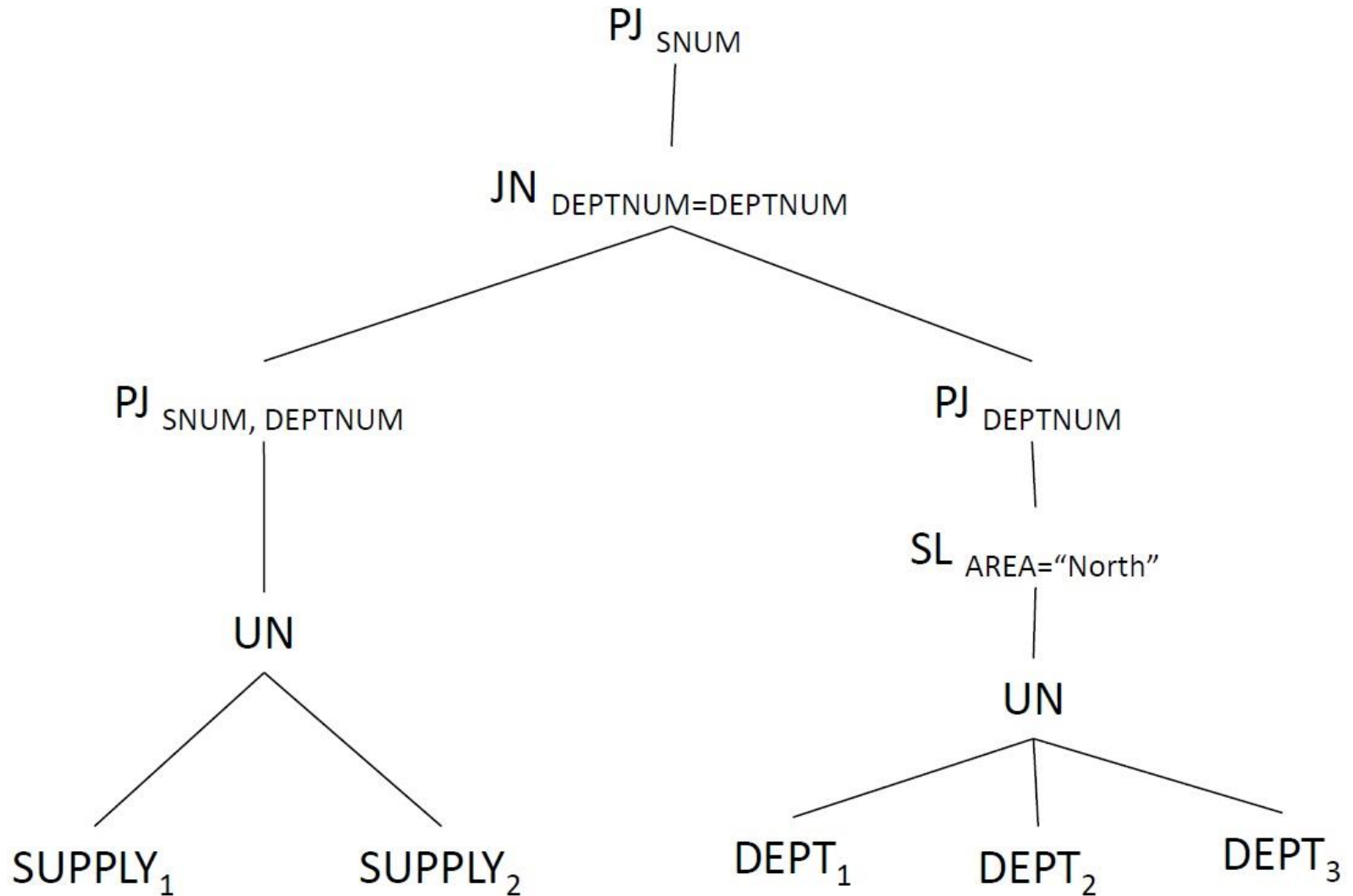
## Input Query (Operator Tree):



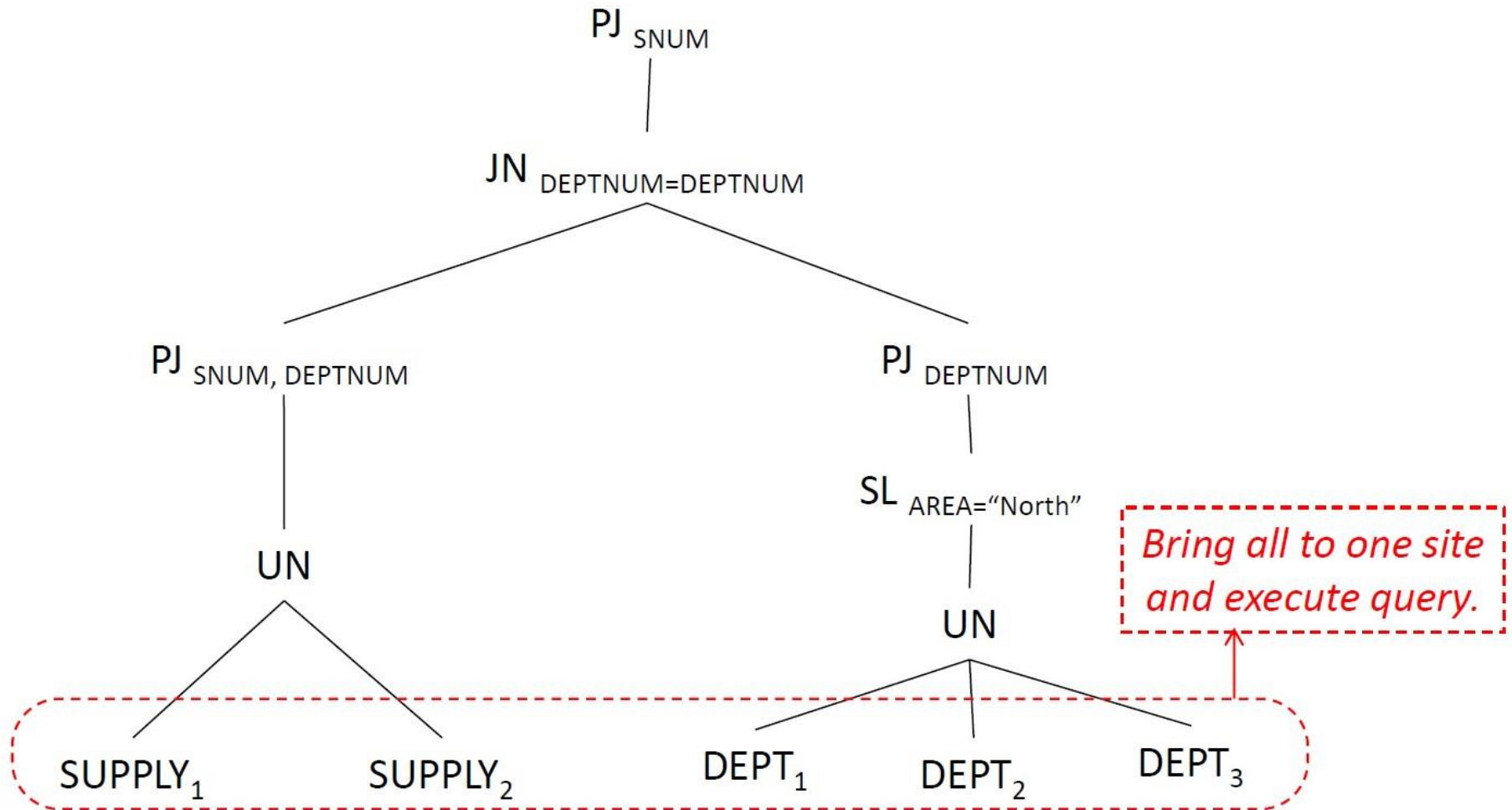
# Strategy – 1

## No simplification

# Strategy – 1



# Strategy – 1 (cont.)



# Strategy – 1 (cont.)

- ❑ Let us execute the query at site 2.
  - We need to collect all the fragments there (in parallel).

# Strategy – 1 (cont.)

- ❑ Let us execute the query at site 2.
  - We need to collect all the fragments there (in parallel).

- ❑ Transmitted amount:

$$X_{\text{SUPPLY}_1} = ?$$

card (SUPPLY<sub>1</sub>) = 30000

|      | snum | pnum | deptnum | quan |
|------|------|------|---------|------|
| size | 6    | 7    | 2       | 10   |
| val  | 1800 | 1000 | 20      | 500  |



# Strategy – 1 (cont.)

❑ Let us execute the query at site 2.

–We need to collect all the fragments there (assume in parallel).

❑ Transmitted amount:

$$\begin{aligned} X_{\text{SUPPLY}_1} &= \text{card}(\text{SUPPLY}_1) \times \text{size}(\text{SUPPLY}_1) \times 8 \text{ bits} \\ &= 30000 \times (6+7+2+10) \times 8 \text{ bits} \\ &= 30000 \times 25 \times 8 \text{ bits} \\ &= 6000000 \text{ bits} \end{aligned}$$

# Strategy – 1 (cont.)

❑ Transmitted amount for other fragments:

$$X_{\text{SUPPLY}_2} = ?$$

$$X_{\text{DEPT}_1} = ?$$

$$X_{\text{DEPT}_2} = ?$$

$$X_{\text{DEPT}_3} = ?$$

card (SUPPLY<sub>2</sub>) = 20000

|      | snum | pnum | deptnum | quan |
|------|------|------|---------|------|
| size | 6    | 7    | 2       | 10   |
| val  | 1800 | 1000 | 20      | 500  |

card (DEPT<sub>1</sub>) = 10

card (DEPT<sub>2</sub>) = card (DEPT<sub>3</sub>) = 20

|      | deptnum | name | area | mgrnum |
|------|---------|------|------|--------|
| size | 2       | 15   | 1    | 7      |
| val  | 10      | 10   | 2    | 10     |

# Strategy – 1 (cont.)

□ Transmitted amount for other fragments:

$$X_{\text{SUPPLY2}} = 20000 \times 25 \times 8 \text{ bits} = 4000000 \text{ bits}$$

$$X_{\text{DEPT1}} = 0 \text{ bits}$$

$$X_{\text{DEPT2}} = 20 \times 25 \times 8 \text{ bits} = 4000 \text{ bits}$$

$$X_{\text{DEPT3}} = 20 \times 25 \times 8 \text{ bits} = 4000 \text{ bits}$$

# Strategy – 1 (cont.)

Assume  $C_0 = 0$  and  $D_0 = 0$

$$\begin{aligned} \text{TC}(x) &= C_0 + (\text{sum of all the amount}) \times C_1 \\ &= (X_{\text{SUPPLY1}} + \dots + X_{\text{DEPT3}}) \times C_1 = 10008000 \times C_1 \end{aligned}$$

$$\begin{aligned} \text{TD}(x) &= D_0 + (\text{largest amount}) \times D_1 \\ &= 6000000 \times D_1 \end{aligned}$$

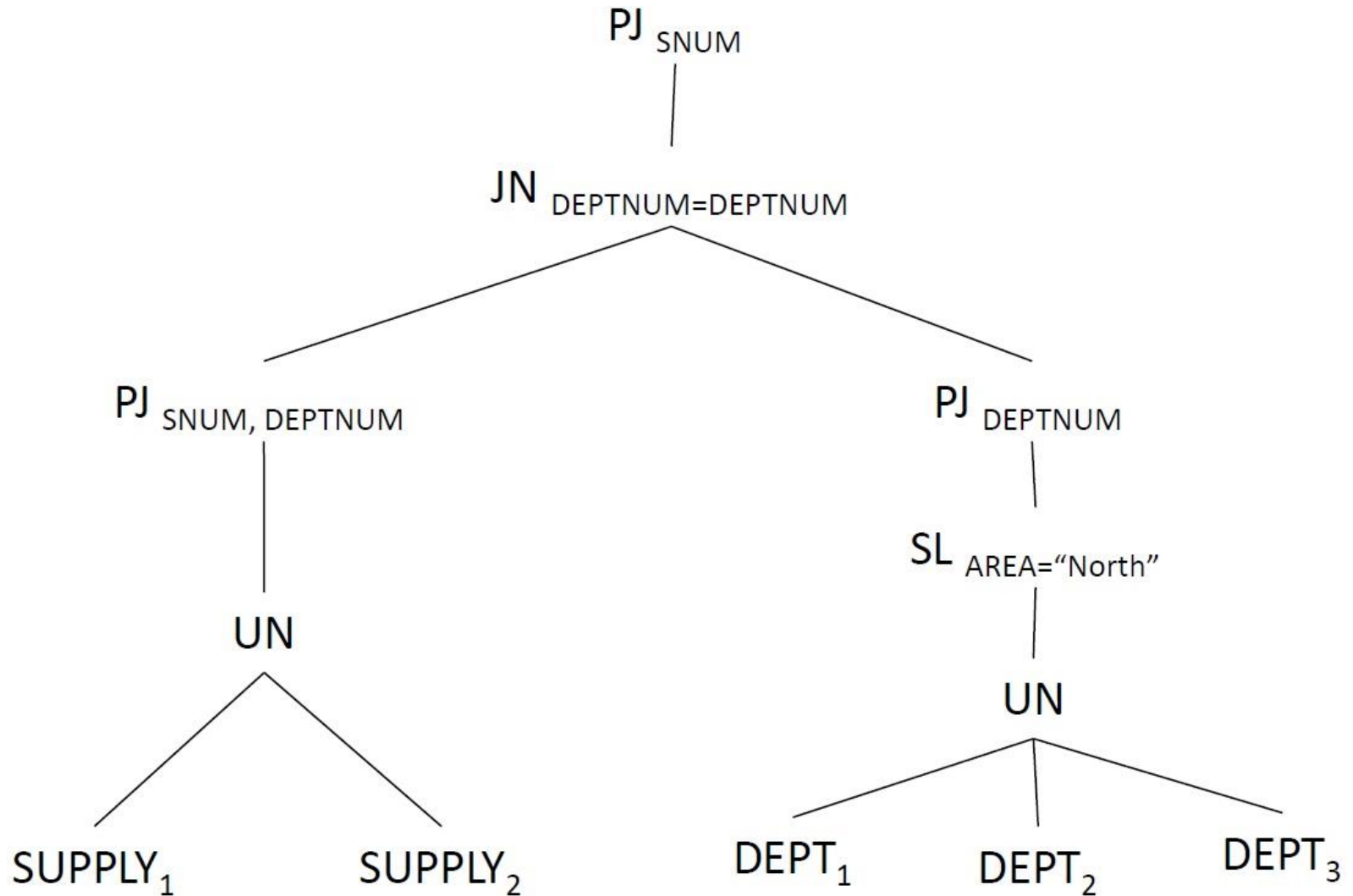
If  $D_1 = 10000$  bit/second,

$$\text{Transmission Delay} = 6000000 \times (1/10000) \text{ s} = 600 \text{ s} = 10 \text{ mins}$$

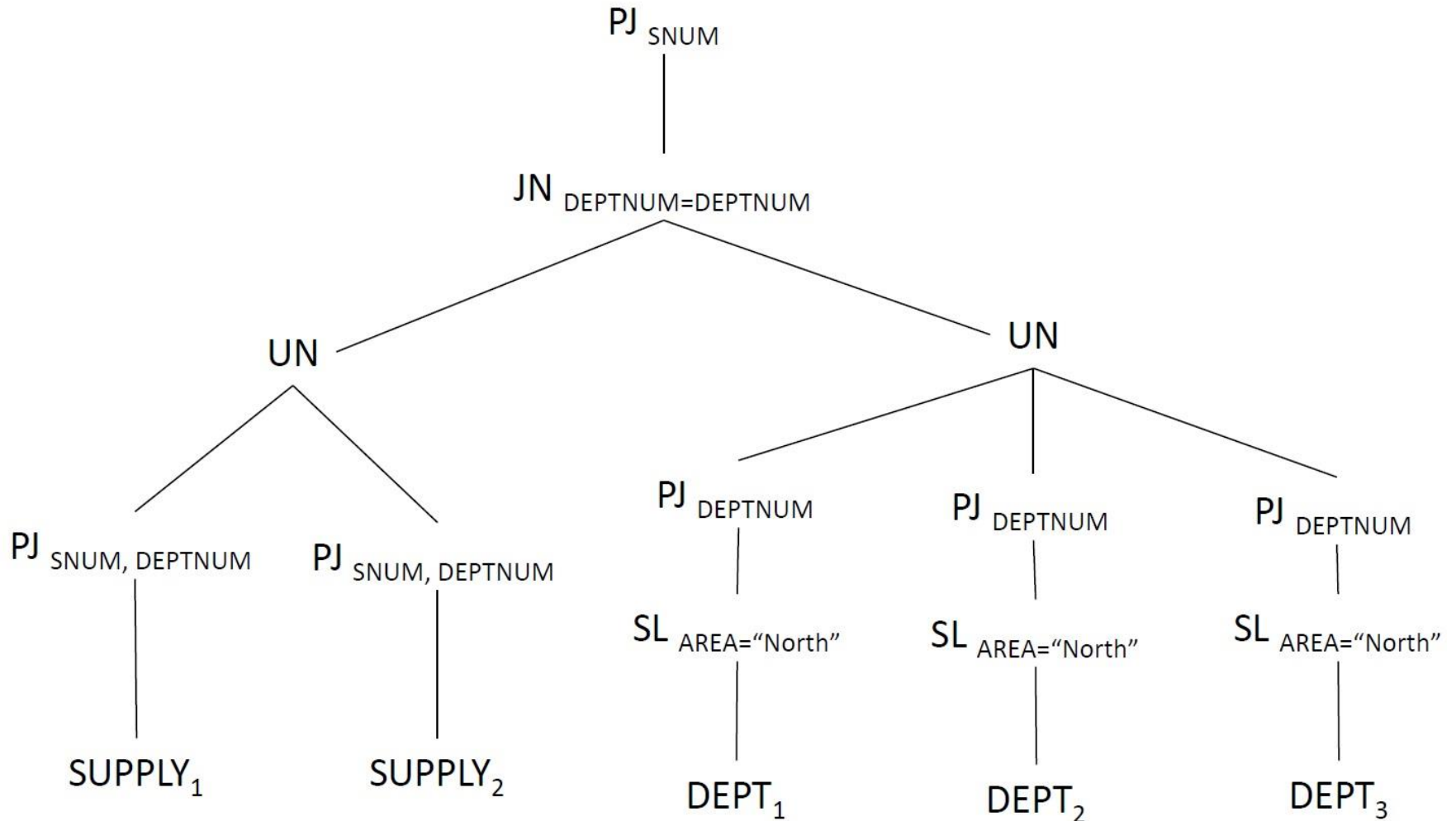
# Strategy – 2

Simplification applied  
(Rules & Criterion -1 and 2).

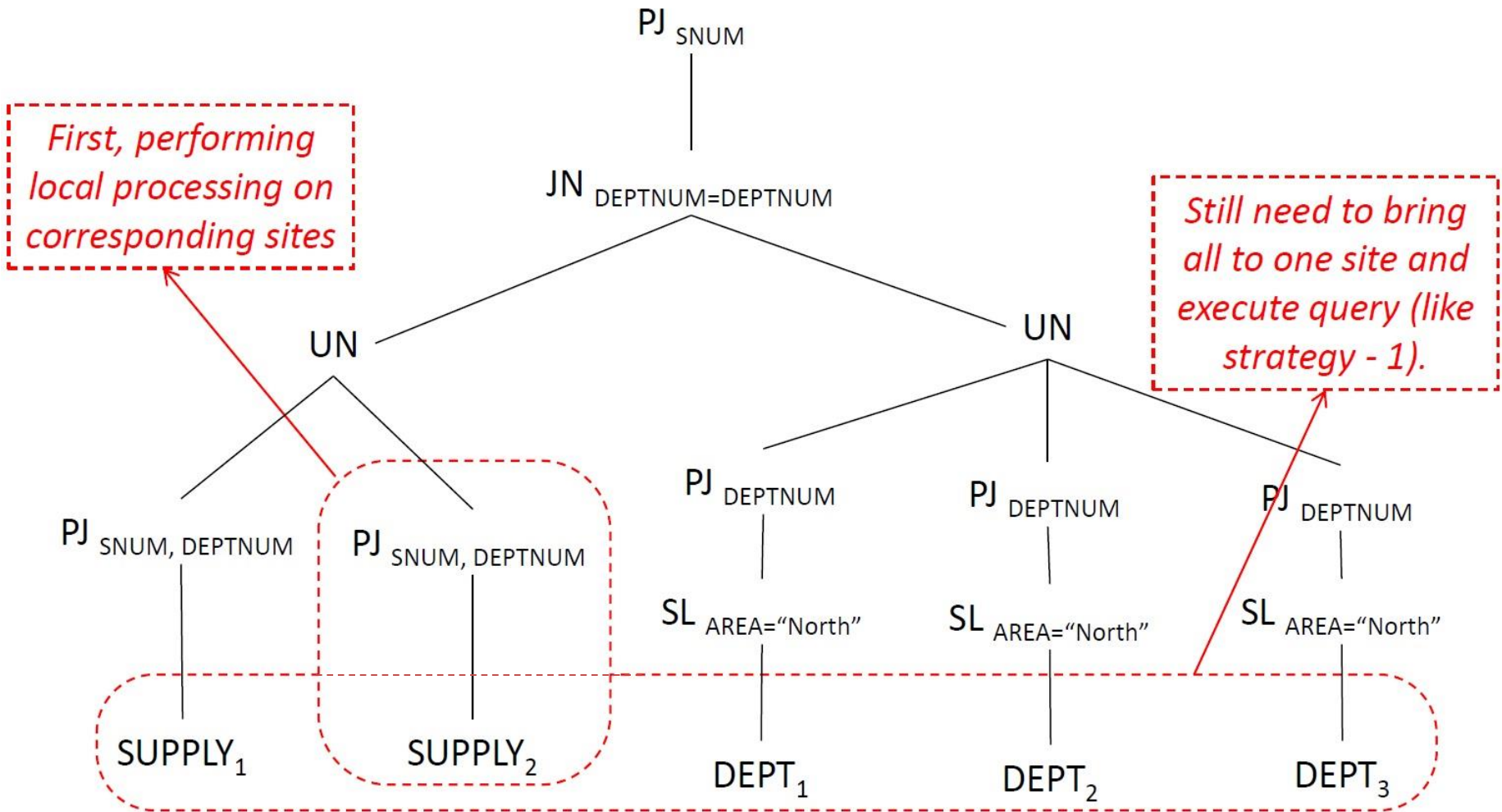
# Strategy – 2



# Strategy - 2



# Strategy – 2 (cont.)





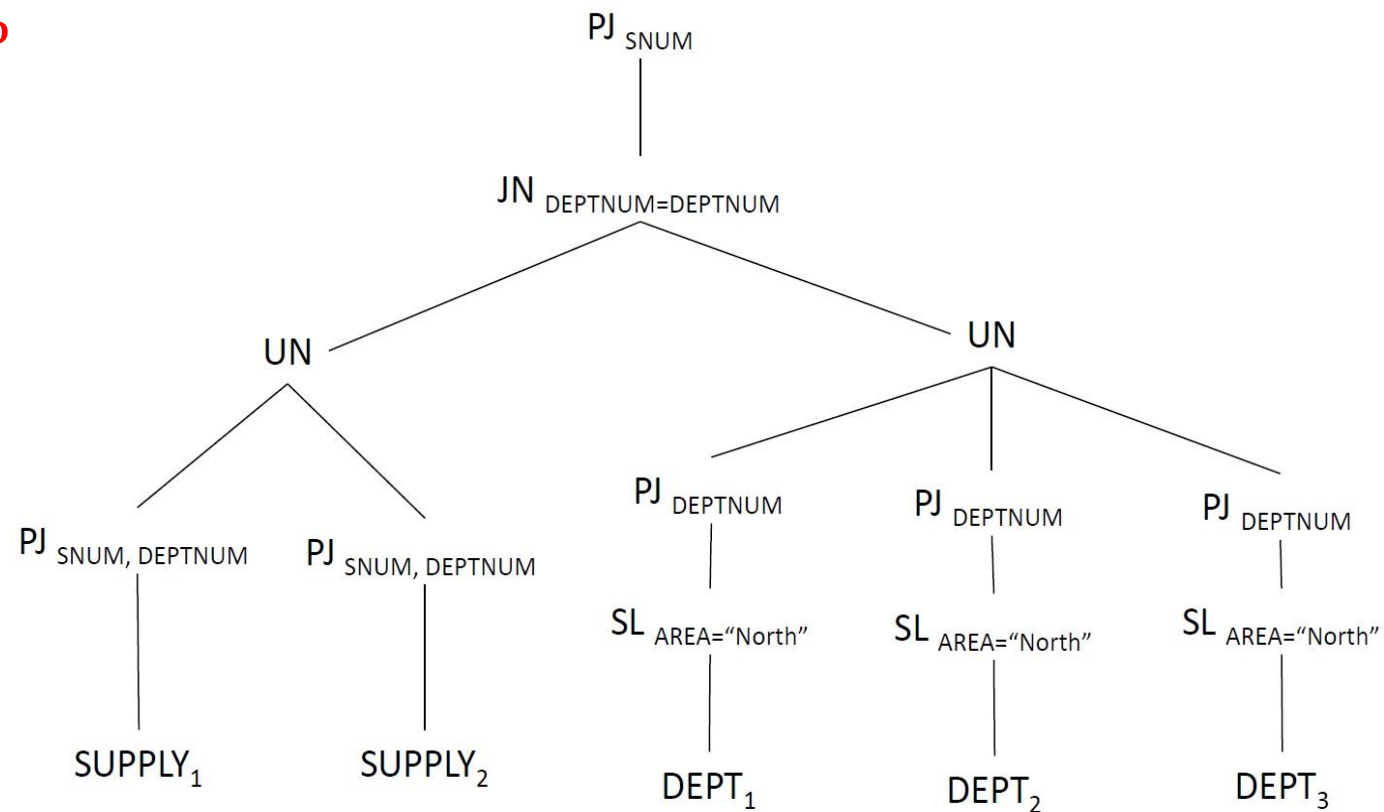
# Strategy – 2 (cont.)

- ❑ Performing local processing on fragments.
  - Fragment reducers.
- ❑ Then sending reduced fragments to the executing site (i.e. site - 2) in parallel.

# Strategy – 2 (cont.)

□ Transmitted amount:

$$X_{\text{SUPPLY}_1} = ?$$



# Strategy – 2 (cont.)

□ Transmitted amount:

$$\begin{aligned} X_{\text{SUPPLY}_1} &= \text{card}(\text{SUPPLY}_1) \times \{\text{size}(\text{snum}) + \text{size}(\text{deptnum})\} \times 8 \text{ bits} \\ &= 30000 \times (6+2) \times 8 \text{ bits} \\ &= 1920000 \text{ bits} \end{aligned}$$

# Strategy – 2 (cont.)

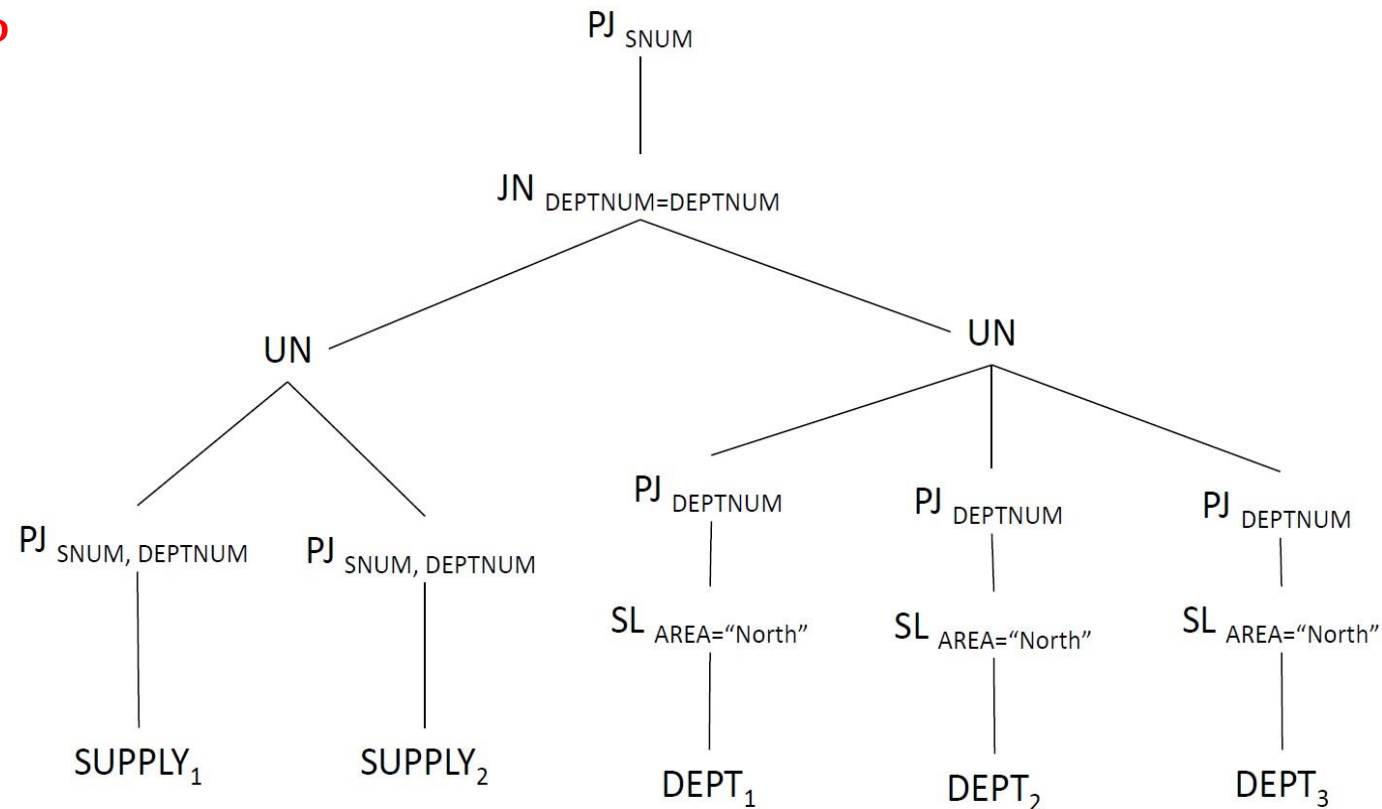
□ Transmitted amount:

$X_{\text{SUPPLY}_2} = ?$

$X_{\text{DEPT}_1} = ?$

$X_{\text{DEPT}_2} = ?$

$X_{\text{DEPT}_3} = ?$



# Strategy – 2 (cont.)

□ Transmitted amount for other fragments :

$$X_{\text{SUPPLY2}} = 20000 \times 8 \times 8 \text{ bits} = 1280000 \text{ bits}$$

$$X_{\text{DEPT1}} = 0 \text{ bits}$$

$X_{\text{DEPT2}} \approx 0 \text{ bits}$  Try to investigate why

$$X_{\text{DEPT3}} = X_{\text{DEPT2}} \approx 0 \text{ bits}$$

card (DEPT<sub>1</sub>) = 10

card (DEPT<sub>2</sub>) = card (DEPT<sub>3</sub>) = 20

|      | deptnum | name | area | mgrnum |
|------|---------|------|------|--------|
| size | 2       | 15   | 1    | 7      |
| val  | 10      | 10   | 2    | 10     |

# Strategy – 2 (cont.)

Assume  $C_0 = 0$  and  $D_0 = 0$

$$\begin{aligned} \text{TC}(x) &= C_0 + (\text{sum of all the amount}) \times C_1 \\ &= 3200000 \times C_1 \end{aligned}$$

$$\begin{aligned} \text{TD}(x) &= D_0 + (\text{largest amount}) \times D_1 \\ &= 1920000 \times D_1 \end{aligned}$$

If  $D_1 = 10000$  bit/second,

$$\text{Transmission Delay} = 1920000 \times (1/10000) \text{ s} = 192 \text{ s} \approx 3 \text{ mins}$$

# Comparisons

| Strategy        | Description                                                                                                                                                                                                                                                           | Time |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1<br>(very bad) | <ul style="list-style-type: none"><li>▪ No simplification, no optimization.</li><li>▪ All fragments are brought to one site to execute the query.</li></ul>                                                                                                           | 10 m |
| 2<br>(bad)      | <ul style="list-style-type: none"><li>▪ Simplification applied (Criterion -1 and 2).</li><li>▪ No optimization.</li><li>▪ Processing on fragments are done on the site locally.</li><li>▪ Then, all fragments are brought to one site to execute the query.</li></ul> | 3 m  |

# Questions

What will happen for *strategy – 1* if the data collection from all the fragments are done sequentially, instead of in parallel?



# Exercise

4. Consider the following global relational schemata.

*EMP* (*ID*, *NAME*, *SAL*, *AGE*, *MGRNUM*, *DEPTNUM*)

*DEPT* (*ID*, *AREA*, *DEPTNUM*, *MGRNUM*)

Corresponding fragmentation schemata:

$EMP_1 = SL_{SAL \leq 25K} EMP$

$EMP_2 = SL_{SAL > 25K} EMP$

$DEPT_1 = SL_{AREA = "North"} DEPT$

$DEPT_2 = SL_{AREA = "South"} DEPT$

Also consider the following global query.

$Q = PJ_{NAME, AREA}(((SL_{SAL > 25K} EMP \Join_{ID=ID} SL_{AREA = "North"} DEPT) \Join_{ID=ID} SL_{AREA = "South"} DEPT) \Join_{ID=ID} SL_{SAL \leq 25K} EMP)$

## Profiles of EMP1 and EMP2:

Card(EMP1) = 1800

Card(EMP2) = 1500

Site(EMP1) = 3

Site(EMP2) = 5

|      | ID   | NAME | SAL | AGE | MGRN. | DEPTN. |
|------|------|------|-----|-----|-------|--------|
| Size | 3    | 4    | 3   | 3   | 2     | 2      |
| Val  | 1000 | 900  | 50  | 30  | 200   | 200    |

## Profiles of DEPT1 and DEPT2:

Card(DEPT1) = 1300

Card(DEPT2) = 2000

Site(DEPT1) = 2

Site(DEPT2) = 4

|      | ID   | AREA | MGRN. | DEPTN. |
|------|------|------|-------|--------|
| Size | 2    | 4    | 2     | 2      |
| Val  | 2000 | 30   | 200   | 200    |

Compare strategy 1 and 2 followed by assembling all the data at **site 3** and execution of the given query  **$Q$** .

Assume that the database profiles of  $EMP_1$  ,  $EMP_2$  are similar. Also, profiles of  $DEPT_1$  and  $DEPT_2$  are identical. Determine the transmission delay for both the strategy when network-wide transfer rate is 20 bytes/second. Note that, the system does not provide any benefit of parallel processing and the initial delay is 0.